So $f(x)$ is an even function.

Hence $$\int_{-\log 2}^{\log 2} e^{-|x|}\,dx = 2\int_{0}^{\log 2} e^{-|x|}\,dx = 2\int_{0}^{\log 2} e^{-x}\,dx = 2(-e^{-x})_0^{\log 2} = 2(-e^{-\log 2}+e^{0}) = 2\left(-e^{\log\frac{1}{2}}+1\right)$$

$$= 2\left(-\frac{1}{2}+1\right) = 1.$$

Example 9.26

Evaluate : $\int_0^a \frac{f(x)}{f(x)+f(a-x)}\,dx.$

Solution

$$\text{Let } I = \int_0^a \frac{f(x)}{f(x)+f(a-x)}\,dx \qquad \ldots (1)$$

Applying the formula $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ in equation (1), we get

$$I = \int_0^a \frac{f(a-x)}{f(a-x)+f(a-(a-x))}\,dx$$

$$= \int_0^a \frac{f(a-x)}{f(x)+f(a-x)}\,dx. \qquad \ldots (2)$$

Adding equations (1) and (2), we get

$$2I = \int_0^a \frac{f(x)}{f(x)+f(a-x)}\,dx + \int_0^a \frac{f(a-x)}{f(x)+f(a-x)}\,dx$$

$$= \int_0^a \frac{f(x)+f(a-x)}{f(x)+f(a-x)}\,dx$$

$$= \int_0^a dx = a.$$

Hence, we get $I = \frac{a}{2}$.

Example 9.27

Prove that $\int_0^{\frac{\pi}{4}} \log(1+\tan x)dx = \frac{\pi}{8}\log 2.$

Solution

$$\text{Let us put } I = \int_0^{\frac{\pi}{4}} \log(1+\tan x)dx \qquad \ldots (1)$$

Applying the property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ in equation (1), we get

$$I = \int_0^{\frac{\pi}{4}} \log\left[1+\tan\left(\frac{\pi}{4}-x\right)\right]dx = \int_0^{\frac{\pi}{4}} \log\left[1+\frac{\tan\frac{\pi}{4}-\tan x}{1+\tan\frac{\pi}{4}\tan x}\right]dx$$

$$= \int_0^{\frac{\pi}{4}} \log\left[1+\frac{1-\tan x}{1+\tan x}\right]dx = \int_0^{\frac{\pi}{4}} \log\left[\frac{1+\tan x+1-\tan x}{1+\tan x}\right]dx$$

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